MILITARY SPECIFICATION

SIGHT, INFINITY: M44C

1. SCOPE

1.1 This specification covers Sight, Infinity: M44C utilized to project a reticle into the field of a gunner's periscope and control the light intensity of the projected reticle.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U. S. Army, Frankford Arsenal

F8565556	Pressure Tester, Two-Station
F8598949	Sight, Infinity: M44C

FSC 1240

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PACKAGING DATA SHEET

8598949 Packaging of Sight, Infinity: M44C

(Copies of specifications, standards, drawings and publications required by supplier's in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- The materials shall be in accordance with the drawings, material specifications, and general specifications forming a part of this specification.

3.2 Fabrication.- The sight shall be manufactured in accordance with Drawing F8598949 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.3 General specifications.- The sight shall meet the following requirements of Specifications MIL-O-13830 and MIL-F-13926.

3.3.1 MIL-O-13830.

(a) Cleanliness
(b) Fixed eyepiece focus
(c) Illuminated reticles

3.3.2 MIL-F-13926.

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.4 Alinement.- With the sight positioned as shown on Drawing F8598951 and with the mounting surface 20 degrees plus or minus 1 minute from the vertical plane and the key horizontal within plus or minus 1 minute, the line of sight from the center of the reticle circle shall be perpendicular to the vertical plane as adjusted by the boresight knobs.

3.4.1 Knob travel.- With the geometric axis as positioned in 3.4, the boresight knobs shall provide at least 10 mils movement in both directions (total excursion of 20 mils) for azimuth and for elevation.

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3.5 Environmental.

3.5.1 Storage temperature.- The sight shall show no evidence of physical failure and shall meet all performance requirements of this specification at a standard ambient temperature of 60° to 90°F. after having been exposed to and thermally stabilized at ambient temperatures of -80° and +160°F.

3.5.2 Operating temperatures.- The sight shall meet the applicable requirements of 3.7 while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.5.3 Vibration.- With the sight mounted in its normal operating position, it shall be vibrated in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds.

3.6 Performance.

3.6.1 Reticle retention.- After being subjected to the requirements of 3.5.3, the line of sight shall not have changed by more than 0.5 mil from the setting established and recorded in 3.4.

3.6.2 Focus.- The reticle ring target shall be focused between 500 - 1000 meters.

3.6.3 Reticle.- The inner diameter of the reticle circle shall subtend an angle of 21 mils ± 3 mils.

3.6.4 Sealing.- The Infinity Sight shall show no evidence of leakage in excess of 0.2 pounds per square inch (psi) when subjected to an internal pressure of 5. ± 0.10 psi for a minimum of 2 hours. The interior atmosphere of the accepted sight shall contain nitrogen having a dew point at least as low as minus 25°F at atmospheric pressure. The sight shall then be sealed with an internal pressure of between 0.25 and 0.50 psig of dry nitrogen.

3.6.5 Illumination.- Utilizing a 24 volt source to illuminate the reticle, the reticle pattern shall appear uniformly illuminated and free of ragged edges or breaks throughout its entire circumference.

3.7 Operability.

3.7.1 Boresight knobs.- The torque required to rotate the azimuth and elevation boresight knobs shall not exceed 7 pound inches at room temperature (plus 60 to 90 degrees Fahrenheit) and at the operating temperature specified in 3.5.2.

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3.7.2 Rheostat.- The torque required to rotate the rheostat knob shall not exceed 4 pound inches, and rotation in both directions shall cause illumination of the reticle to vary appropriately when checked at the standard ambient temperature (plus 60° to 90°F) and the operating temperatures specified in 3.5.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 sights. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

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4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein, shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS

CLASS	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 0.65% Defective		
101. Vibration	3.5.3	4.6.5
102. Alinement	3.4	4.6.1
103. Knob travel	3.4.1	4.6.2
104. Reticle retention	3.6.1	4.6.6
105. Focus of reticle	3.6.2	4.6.7
106. Reticle	3.6.3	4.6.8
107. Illumination of reticle	3.6.5	4.6.10
108. Boresight knobs operability	3.7.1	4.6.11
109. Rheostat operability	3.7.2	4.6.12
110. Sealing	3.6.4	4.6.9
MINOR: NONE		

NOTE: The tests in Table I shall be conducted at a temperature between 60° and 90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

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4.4.1 General.- One sight shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Drawings, Specifications - Visual
302.	Fabrication	3.2	Drawing - Visual
303.	General specifications	3.3	MIL-O-13830, MIL-F-13926 - Visual

NOTE: The tests in Table II shall be conducted at a temperature between 60° and 90°F.

4.4.2 Environmental.- Three sights shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall meet the requirements and tests in Table III and shall then meet the requirements for characteristics 108 and 109 in Table I.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
304.	Storage temperature (-80°F and +160°F)	3.5.1	4.6.3
305.	Boresight knobs operability (-40° and +150°F)	3.5.2, 3.7.1	4.6.4, 4.6.11
306.	Rheostat operability (-40° and +150°F)	3.5.2, 3.7.2	4.6.4, 4.6.12

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by

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drawing number in specifications or SQAP's forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-P-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Equipment to test for environmental requirements.- All environmental testing equipment used to determine conformance with the temperature and vibration requirements shall conform to the "Test Facilities" requirements contained in Specification MIL-P-13926.

4.5.2.3.2 Equipment to test for performance requirements.- The special testing equipment shall position the infinity sight on a keyed locating surface with the mounting surface 20 degrees plus or minus 1 minute from the vertical plane and the key horizontal. The test equipment shall be equipped with a medium capable of measuring the projected reticle image of the infinity sight, in increments of 1 mil within an accuracy of plus or minus 0.2 mil in both horizontal and vertical directions, in a manner suitable for measuring the anglar subtense of the projected reticle pattern, and boresight adjusting requirements. The equipment shall be capable of relocating the infinity sight within 0.05 mil of the original mounting position. Provisions shall be made to adjust the measuring medium with respect to the test equipment mounting surface and the horizontal locating key to an angular accuracy of 1 minute respectively.

4.5.2.3.3 Boresight knob.- The equipment shall consist of a torque wrench type tester which will permit reading in both the clockwise and counterclockwise directions with a minimum range of from zero to ten pound inches, the accuracy of which shall be within plus or minus one-quarter pound inch throughout the entire zone of excursion. It shall be equipped with an adapter to accommodate the boresight knob.

4.5.2.3.4 Rheostat knob.- This equipment shall consist of a torque wrench type tester which meets the requirements specified in 4.5.2.3.3 and shall be equipped with an adapter to accommodate the rheostat knob.

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4.5.2.3.5 Equipment to test sealing.- The special testing equipment to perform the internal pressure test shall conform to Drawing F8565556, Pressure Tester-Two Station, along with a standard commercial type dew point tester and regulator.

4.6 Test methods and procedures.

4.6.1 Alinement.- The infinity sight shall be positioned and alined using special test equipment as defined in 4.5.2.3.2 to hold the sight as specified in 3.4. An auxiliary telescope (projected collimator) with an attached eyepiece will aid in measuring the projected reticle image of the infinity sight. The collimator used shall contain a reticle with either a mil grid pattern or cross lines graduated in mils spaced about the geometric axis of the collimator target. Connect a 24 volt power supply to the contact plug of the sight. Rotate variable resistor knob clockwise to illuminate reticle. The line of sight through the center of the illuminated reticle shall be directed to the reticle of the collimator, using the boresight knobs to center the image on the mil grid of the collimator.

4.6.2 Knob travel.- Conformance with the requirements of 3.4.1 will be determined by visual and manual examination, with the sight positioned as in 4.6.1 and the reticle in focus in the central position. Observe upper edge of the illuminated reticle. Rotate elevation boresight knob counterclockwise until the upper edge of the sight reticle reads at least 10 mils in movement from its central position. Return circle to center position by clockwise rotation of knob. While observing the lower edge of the illuminated circle and clockwise rotation of knob, bring the circle down until the lower edge reads at least 10 mils in movement. Return reticle to central position. Repeat this procedure with the azimuth boresight knob. Range of travel in azimuth shall move the reticle image at least 10 mils right and 10 mils left of center.

4.6.3 Storage temperatures.- The sights shall be exposed to the temperatures indicated in 3.5.1 utilizing test facilities conforming with the requirements of 4.5.2.3.1. The sights shall be placed in the test chamber and the temperature of the chamber reduced gradually to -80°F and shall remain at this temperature for a minimum of five hours. At the completion of this five hour period, the temperature shall then be gradually raised to +160°F and the sights shall remain at this temperature for a minimum of five hours. At the completion of this five hour period, the temperature of the test chamber shall be gradually reduced to room ambient temperature (+60° to +90°F) at which time the sights shall be removed from the test chamber. The sights shall be examined externally and internally by visual and tactile means to determine that there is no evidence of physical defects. Caution shall be exercised during this temperature test to avoid subjecting the infinity sight to thermal shock. The sights shall then be subjected to the performance tests specified herein to determine overall compliance with 3.5.1.

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4.6.4 Operating temperatures.- The sights shall be exposed to the temperature conditions specified in 3.5.2 employing the same test facilities used for the temperature test of 4.6.3. The sights shall be placed in the test chamber and the temperature of the chamber reduced gradually to -40°F. The sights shall remain at this temperature for a minimum of 5 hours. At the completion of this 5 hour period, and while at -40°F the sights shall be subjected to the tests in 4.6.11 and 4.6.12 to determine compliance with 3.5.2, 3.7.1 and 3.7.2. Upon completion of these tests the temperature of the test chamber shall be raised gradually to +150°F. The sights shall remain at this temperature for a period of 5 hours and at the conclusion of this 5 hour period (while at +150°F) the sights shall again be subjected to the tests in 4.6.11 and 4.6.12. Upon completion of this test the temperature of the test chamber shall be gradually reduced to room ambient temperature of 60° to 90°F at which time the sights shall be removed from the test chamber. Caution shall be exercised during this temperature test to avoid subjecting the sights to thermal shock.

NOTE: The rate of temperature change in the test chamber shall not exceed 20°F per hour throughout the temperature cycling test of 4.6.3 and 4.6.4.

4.6.5 Vibration.- Standard vibrating equipment capable of providing the amplitude and frequency as specified in 3.5.3 shall be utilized for the vibration test. Unless otherwise specified, the test facilities shall conform with the requirements of 4.5.2.3.1. A fabricated adapter shall be provided that will hold the sight in its normal operating position and will assure a positive and secure mounting of the sight on the vibration machine. After inspection for boresight knob travel, as outlined in 4.6.2, care shall be exercised not to disturb the settings of the boresight knobs when removing the sight from the special test equipment. Each infinity sight in the sample shall be secured to the vibrating table in the position specified in 3.5.3 by means of the fabricated adapter, and vibrated to the requirements of 3.5.3. At the conclusion of this vibration remove the sight from the vibration machine and examine the sight externally and internally for any evidence of physical failure. A visual examination shall be made for missing or damaged parts, with particular attention to the optical elements of the sight. A tactile examination shall be performed to determine looseness or improper function of moving parts, such as the variable resistor control knob. Sights that have passed the above examinations shall then be remounted on the special test equipment as in 4.6.1. During this phase, care shall be exercised not to disturb the boresight knobs setting.

4.6.6 Reticle retention.- Subsequent to the test of 4.6.5 the line of sight shall be checked again in accordance with 4.6.1. The position of the line of sight shall be identical to the original setting in 4.6.1. Any difference in these settings, resulting from the vibration test, shall not exceed the value specified in 3.6.1.

4.6.7 Focus.- The objective of the auxiliary telescope (collimator) shall be set between 500 and 1000 meters, and the collimator eyepiece

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adjusted. Centralize the reticle image of the sight unit under test by movement of the boresight knobs. Observe the image of the sight reticle formed on the collimator reticle. The sight reticle image must be in sharp focus on the collimator reticle to comply with the requirements specified in 3.6.2.

4.6.8 Reticle.- The test for the angular value of the reticle target ring is accomplished by checking it against the collimator containing a reticle target with dimensions suitable for the target distance utilized, and based on the reticle ring dimensions shown on the drawing. The error for this test shall not exceed the limit specified in 3.6.3.

4.6.9 Sealing.- Internal pressurizing of the sight shall be performed with special testing equipment conforming to 4.5.2.3.5. Pressurize the sight with dry nitrogen gas, having a dew point at least as low as minus 25°F, to 5 pounds per square inch (psi) in accordance with the procedure for internal testing on Drawing F8565556. Retain the applied pressure in the sight for the time duration specified in 3.6.4. During the first 50 minutes of this time period, the elevation and azimuth knobs shall be rotated to their clockwise and counterclockwise extremes, a minimum of 3 times. At the end of the 2 hour period there shall be no evidence of leakage as indicated on the pressure gage of the testing equipment, to determine compliance with 3.6.4. Upon satisfactory completion of the sealing test, the sight shall be flushed with the dry nitrogen gas specified herein. The interior atmosphere of the sight shall be periodically sampled through a dew point meter until the atmosphere shows a dew point at least as low as minus 25°F. When this atmosphere is attained within the sight, the sight shall then be sealed in accordance with 3.6.4. The interior atmosphere of the infinity sight shall conform to the requirements of 3.6.4.

4.6.10 Illumination.- Illuminate the reticle lamp by applying the voltage specified in 3.6.5 through a connection connected to the contact plug on the housing assembly. The reticle pattern shall be visually observed to determine compliance with the requirements of 3.6.5. Clockwise rotation of the rheostat knob shall cause the illumination of the reticle to get brighter and counterclockwise rotation shall dim the reticle illumination.

4.6.11 Boresight knob operability.- The running torque required to rotate the specified knobs shall be measured using test equipment that meets the requirements of 4.5.2.3.3. Torque shall be measured while rotating the knobs through its full range of movement. Torque values obtained at minus 40 degrees and plus 150 degrees Fahrenheit shall be in conformance with 3.7.1.

4.6.12 Rheostat operability.- The running torque required to rotate the rheostat knob shall be measured using test equipment that meets the

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requirements of 4.5.2.3.4. Torque shall be measured while rotating the knob through its range of movement in both directions, and illumination of the reticle shall be visually observed to determine compliance with the requirements specified in 3.7.2.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 8598949. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Infinity Sight: M44C is used in conjunction with Periscope, M31 to superimpose an illuminated reticle circle upon the target as seen through the periscope unity power optical system and serves as a sighting reference for the coaxial machine gun.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging and packing.
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.
(e) Requirement for first article testing.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1240-A295

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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